

[Dr. Vijay Gokhale Sir]

Books:

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The C Programming Language:

Beginner:

- *[1] Programming with C By Gottfried (Schaum's Series)
- [2] Programming in C (Ajay Mittal)

Building expertise:

- *[1] Understanding and Using C Pointers - Richard Reese (O'Reilly)
- [2] Expert C Programming Deep C secrets - Peter van den Linden

Post Expert level:

- *[1] The C Programming Language by K & R [Brian Kernighan & Dennis Ritchie]
- [2] GCC User Manual manual : Chapter 5/6 -> Extensions to C language family
(after reading this chapter you will be eligible for kernel programming from language perspective)

Recommended Track:

- [1] Gottfried [2] Understanding & Using C Pointers [3] K & R [4] GNU C manual

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Assembly Programming Language:

Linux :

- *[1] Intel 32 bit assembly programming + Linux OS + GNU tool-chain
Professional Assembly Language by Richard Blum
- [2] Assembly Language : Step by Step : Programming with Linux
Duntemaan
- [3] Modern x86 assembly language programming Kusswurm
- [4] Beginning x64 assembly programming Appress.

Windows:

- *[1] Assembly programming for intel based programmer
x86 assembly programming : Kip Irvine (7th Edition)

Historic:

- *[1] IBM PC Language -> Peter Abel
- *[2] Advanced MS DOS Programming -> Ray Duncan

Manual:

- *[1] Intel Software Developers Manual Vol 1
- *[2] Intel Software Developers Manual Vol 4 : Optimization Manual

Protected Mode Features (assembly for OS internals)

- [1] Advanced 80386 Programming by James Turly
- [2] -----"----- MS Press (VERY RARE)
- **[3] x86 Instruction Set Architecture -> By Tom Shanley
- [4] Intel Software Developers Manual Vol 3A - 3B

Hardware Track

- *[1] ISA system bus architecture
- [2] EISA system bus architecture
- [3] 80486 architecture
- [4] Protected Mode Architecture

x[5] PCI system bus architecture

x[6] USB



[7] Serial Port Complete / Parallel Port Complete / UBS Complete
Jack Achelson

Track:

[1] Richard Blum [2] Kip Irvine [3] Intel Manual vol 4

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The C++ Programming Language -> The Queen of All Programming Language

Quick Start:

- [1] C++ Primer Plus
- [2] C++ : How to Program
- [3] Thinking in C++ (vol 1 , Vol 2)

Beginner

- [1] Essential C++ -> Stanley Liepmann
- *[2] C++ Programming : Principles & Practice -> Bjarne Stroustrup
(1st edition : 98, 2nd edition : 11, 14)

Intermediate

- *[1] C++ Primer : Stanley Liepmann

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3rd edition : 98 C++ : best for knowing how C++ compiler does things internally
4th edition : 98 C++ : readable (3rd edition is not readable unless you are

already familiar with C++)

5th edition : C++ 11, 14 preview

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Advanced:

- [1] Effective C++ --> 1st edition (98)
- [2] More Effective C++ --> 1st edition (98)
- *[3] C++ Template Programming : In depth Guide (1st edition 98 C++)
-> Jossuets & Van der Voorde
(2nd edition : 2017 C++ : Template)
- [4] The C++ Programming Language : Bjarne Stroustrup
[3rd edition : 98 C++
4th edition : 11 C++]

Standard Template Library:

- [1] Basic : Designing Components with C++ STL : Addison Wesley
- *[2] Advanced & Bible : The C++ Template Library : Jossuets
[1st edition : 98, 2nd edition : 2017 STL]
Not only, 'How to use STL' but STL implementation details (C++ template in depth

guide

pre-requisite)

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Internals & Design:

- *[1] Inside C++ Object Model By Liepmann
- *[2] Design & Evolution of C++ -> Stroustrup

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Zen Level:

- *[1] Object Oriented C -> Alex Tobius Schneider

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Track:

- [1] C++ Programming : Principles & Practice -> Bjarne Stroustrup
- [2] C++ Primer [need not be read page to page, 3rd & 4th]
- [3] C++ Template Programming : In depth Guide (1st edition 98 C++)-> Jossuets & Van der Voorde
(2nd edition : 2017 C++ : Template)
- [4] The C++ Programming Language -> Stroustrup (3rd and / or 4th)(wont need line by line)



[advices, part 4 : design, chapter 2/3 : A tour of C++,
Booklet : A tour C++ -> book]

[5] Advanced & Bible : The C++ Template Library : Jossuits

[1st edition : 98, 2nd edition : 2017 STL]

Not only, 'How to use STL' but STL implementation details (C++ template in depth guide
pre-requisite)

[6] Inside C++ Object Model By Liepmann

[7] Design & Evolution of C++ -> Stroustrup (at leisure)

[8] Object Oriented C -> Alex Tobius Schneider

Will be explained later

[9] Inside COM -> Del Rogerson

[10] Essential COM -> Don Box

[11] Inside OLE 2.0 -> Kraig Brockshmidt

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The Java Programming Language:

Core Java

*[1] Core Java Vol 1 by Cay Horstman & Horstman (11th Edition)

*[2] OCJP Java : Khalid Mughal (cannot be read beginners)

Advanced Java

[1] Core Java Vol 2: Cay Hosrtman & Horstman

Java Enterprise API's

[1] Oracle Vol 1 & Vol 2 (freely available on oracle webside)

Prerequisite : Design Patterns in Java (Pearson)

Java Internals:

*[1] Java Virtual Machine specification (Documentation)

*[2] Java Performance Tuning (Pearson) -> James Gosling

Framework:

Documentation

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C#

[1] CLR via C# -> Jeffrey Richter

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Essential .Net Volume 1 - Don Box

Dot net Book Zero : By Charles Petzold

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Perl

[1] Learning Perl Oreilly

*[2] Programming Perl-> Larry Walls (bible)

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Python:

*[1] Learning Python - Mark Lutz (1500)

*[2] Programming Python - Mark Lutz (1700)

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Haskell :

**[1] Haskell road to logic, maths, and programming

**[2] Discrete mathematics using computers

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Golang

[1] The way to Go by Ivo Balbaert



*[2] Concurrency in Go by Katherine Cox - Buday
[3] The Go Programming language -> Kernighan & Pike

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Data Structures and algorithms!

Language Agnostic

**[1] Introduction To Algorithms by Cormen, Rivest, Leiserson -> Valley Interview Book

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Introduction to Discrete mathematics with applications to CS -> Kenneth
first 7 chapters

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(not to be read right now!)

C

[1] Fundamentals of Data Structure through C -> Horowitz Sahani

[2] Algorithms in C -> Vol 1-4 (Book 1), Vol 5(Book 2)

C++

*[1] Data Structure & Algorithms by Tamassia & Goodrich

[2] Data Structures and Problem Solving using C++ -> Mark Allen Weiss

Java

[1] Algorithms in Java -> Robert Sedgewick

[2] Data Structure & Algorithms (java edition) -> Tamassia & Goodrich

**[3] A practical guide to data structure & algorithms using Java Goldman & Goldman

Python

[1] Data Structure & Algorithms (Python edition) -> Tamassia & Goodrich

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The Art of Computer Programming Vol 1 to Vol 4 -> Donald E. Knuth.

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Operating System and Internals:

Admin

Linux:

- [1] Running Linux
- [2] Mastering Shell Scripting. Wrox Publication
- [3] Advanced bash scripting -> LDP
- [4] linuxfromscratch.org

System Call Programming

Pre-requisite -> C Programming Language

Unix/Linux/POSIX:

- [1] Advanced Programming Under Unix Environment W Richard Stevens / Rago -> 3rd

Edition

- **[2] The Linux Programming Interface - Michael Kerrisk

Windows:

- *[1] Programming Windows -> Charles Petzold
- **[2] Programming Applications for MS Windows (4th Edition)-> Jeffrey Richter
- *[3] Programming Server Side Applications for MS windows -> Jeffrey Richter (Jeevan)
- [4] Windows System Programming -> Hart

Device Drivers:

Linux:

- [1] Linux Kernel Module Programming Guide
- *[2] Linux Device Drivers by Rubini & Rubini



[3] Essential Linux Device Drivers

Windows:

- **[1] Viscarola & Mason
- *[2] Art Baker
- *[3] Walter Oney
- *[4] Windows Drivers Foundations -> Microsoft Publication
- [5] Windows kernel programming Yosifovich

Unix:

- [1] Unix Device Drivers : Pajari

O.S. Internals:

Generic Concept (O.S. Agnostic)

- *[1] Modern Operating System by A.S. Tanenbaum
[Chapter 2 : The Process : Busy Waiting -> Monitors]
- [2] William Stalling -> Operating systems

OS Specific Design

- **[1] The Design of Unix OS -> Maurice J Bach
- **[2] Magic Garden Explained -> Cox & Goodhart (Unix Sys V R4)
- **[3] The Design and Implementation of 4.4 BSD -> McKusick & McKusick
The Design and Implementaiton of FreeBSD -> --||-----
(Research Paper : Before Memory was Virtual -> PJ Denning)
- **[4] Solaris Internals -> Jim Mauro & McDougal
1st and 2nd Edition [Kthread -> LWP -> Uthread]
[POSIX thread Library -> m : 1 to m : n to 1 : 1]
- **[5] Unix Internals : New Frontiers -> Vahalia
[Process Model : Unix Process -> Multi-threaded Evolution]
- **[6] Inside Windows NT -> Helen Custor 1st
Inside Windows NT -> David Solomon 2nd
Inside Windows 2000 -> David Solomon 3rd
Windows Internals 4th -> David Solomon & Mark Russinovich
(7th Edition)
- **[7] Inside MAC OS X -> Amit Singh

OS Specific Implementation

- [1] Linux Kernel Development by Robert Love

OS Specific Implementation with Hardware specific details

- **[2] Understanding the Linux Kernel by Cessati & Bovet
- [3] Linux Kernel Architecture by Maurer (2.6.24)

Kernel Code Commentary & Subsystem Specific Books!

- **[1] John Lion's Kernel Code Commentary (UNIX version 6)
- **[2] The Design and Implementation of Operating System : MINIX 3 Book.
A.S. Tanenbaum.
- **[3] Maxwell's Kernel Code Commentary -> 2.4.x Kernel
- **[4] Understanding Linux VM Manager -> Mel Gorman (2.4.22)
- **[5] Understanding Linux Process Manager -> Gorman (2.4.20)
- **[6] What is Page?

Embedded Linux:

- [1] Building Embedded Linux
- **[2] Exploring Beaglebone
- [3] Exploring Rpi
- [4] Linux kernel In Nutshell



[5] Embedded Linux Primer

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*****PRECALCULUS MATHS*****

Mathematics

- [1] 5-10 standard CBSE or ICSE board books [starter]
- [2] Algebra and Trigonometry by Sullevin (10th edition)

Basic Algebra :

- [1] Higher Algebra -> Hall & Knight
- [2] Higher Algebra -> Bernard & Child

Euclidean Geometry:

- [1] Euclidean Geometry -> karmalkar
- [2] Maths Olympiad Geometry books

Trigonometry:

- [1] Plane Trigonometry vol 1 -> SL Loney
- [2] Plane Trigonometry vol 2 -> SL Loney
- [3] Maths Olympiad books

Coordinate Geometry

- [1] 2D coordinate geometry by SL Loney

*****Discrete mathematics*****

- **[1] DM with applications to CS -> Rosen
- [2] Discrete and Combinatorial Maths -> Grimaldi & Ramana
- *[3] How to prove it? Velleman (Cambridge)
- [4] Introduction to Set theory -> Halmos
- [5] The elements of set theory (!)
- **[6] Logic, Proofs and Languages -> STANFORD UNIVERSITY BOOK
- [7] Applied Combinatorics -> Alan Tucker
- [8] Graph Theory -> Diestel (Graduate) (VA)
- x[9] Graph Theory -> Harary (VA)
- x[10] Graph Theory -> Bondy & Murty (VA)
- x[11] Abstract Algebra -> Herstein
- x[12] Abstract Algebra -> Dummit & Foote
- [13] Elementary Number Theory -> David Burton
- x[14] Number Theory -> G.H. Hardy

- [1] Foundations of Discrete Mathematics -> K.D. Joshi
- [2] Applied Discrete Structures -> K.D. Joshi

*****Calculus*****

- **[1] Thomas Calculus
- **[2] Calculus -> Early Transcendentals -> Stewart
- [3] Problems in Calculus of One Variable -> I.A. Maron
- [4] Advanced Engineering Mathematics (Differential Equation)
- [5] Fourier Transform / Laplace's Transform / DE -> Schaum's series

*****Analysis*****

- *[1] Introduction to real analysis -> Sherbert
- [2] Principles of mathematical analysis -> Walter Rudin
- [3] Functional analysis -> Walter Rudin
- [4] Real & Complex Analysis -> Walter Rudin
- [5] Real Analysis -> Royden
- [6] Calculus for scientists & Engineers (Calculus + Real analysis)
- [7] General Topology -> Munkres
- [8] Introduction To General Topology -> K.D. Joshi

Physics

- [1] Haliday & Resnick
- [2] University Physics (Sears and Zemansky)



- [3] Problems in General Physics I.E. Irodov (Classical Mechanics & Electromagnetism)
- [4] Lectures in Physics (vol 1 to vol 3 -> Richard Feynman)

Semiconductor Physics

- [1] Semiconductor Physics -> Donald Naeman (Semiconductor analysis at Quantum level)

Electronics Devices and Circuits

- [1] Yellow Milmaan
- [2] Black Milmaan
- *[3] Electronics Devices & Circuits -> Donald Naeman

Digital Electronics

- [1] R.P. Jain
- [2] Fundamentals of Digital Design with Verilog -> Stephen Brown
- [3] Fundamentals of Digital Design with VHDL -> Stephen Brown

Computer Organization

- *[1] Computer Organization -> A hardware / software interface -> Hanessey & Patterson
- *[2] Computer Architecture -> A Quantitative Approach -> Patterson & Hanessey
- [3] CO -> Zaakey (Good text book)
- [4] CO -> Hays (Good book)
- *[5] Structured Computer Organization by AST
- *[6] Computer Organization & Design -> P. Pal Chaudhary

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TOC & Compilers & Type Theory

TOC

- *[1] Introduction to automata theory, languages and computation
Aho Ulmaan Hopcraft
- *[2] Introduction to Theory of Computation -> Martin

Compilers

- *[1] Compilers : Principles, Techniques & Tools aka (Dragon Book)
Aho Ulmaan Sethi

- **[2] Theory of parsing, translation & compiling -> VOL 1 -> Aho Ulmaan

- [3] Engineering a compiler

- [4] Crafting a compiler

Types Theory and lambda calculus

- [1] Types & Programming Languages -> Benjamin Pierce
- [2] Advanced Topics in Types and Programming Languages -> Benjamin Pierce
- [3] Design Concepts in Programming Languages by Sheldon

Pre->

- [1] Principles of Programming Languages by Ravi Sethi
- [2] Programming Language Pragmatics -> Michael Scott

OOP concepts & design pattern

- *[1] Object Oriented Analysis and Design by Grady Booch -> 2nd Edition / 3rd Edition
- *[2] Object Oriented Software Construction by Bertrand Meyer
- *[3] Design Patterns : An elements of reusable components

Thoery if you want to try practical COM

- [1] Inside COM -> Del Rogerson
- [2] Essential COM -> Don Box
- [3] Inside OLE 2.0 -> Kraig Brockschmidt



Database

- [1] Database system concepts by Korth 7th edition
- [2] Evan Byrass
- [3] DBMS -> Navathe

Networking

General Theory

- [1] Networking -> Top to Down approach -> Forouzan

TCP / IP Theory

- *[1] TCP/IP Illustrated Vol 1 -> W Richard Stevens
- *[2] Internetworking with TCP / IP -> Vol 1 -> Douglas Comer
- *[3] The TCP/IP Guide

Programming

- **[1] Unix Network Programming Vol 1 : 3rd
- [2] Unix Network Programming Vol 2 : IPC
- [3] Unix Network Programming Vol 3 : HTTPS etc. (C programming)
- [4] Internetworking with TCP / IP : Vol 3 : BSD Sockets
- [5] Windows Socket Programming [MS Press]

Internals

- [1] TCP/IP Illustrated Vol 2 : W Richard Stevens (4.4 BSDlite)
- [2] Internetworking with TCP/IP -> Vol 2 -> XINU
- [3] Understanding Linux Network Internals

Distributed Systems

- [1] Distributed Systems -> Principles & Paradigms (Tanenbaum & Van Der Steen)
- [2] Distributed OS -> Tanenbaum

AI

- **[1] Artificial Intelligence : A Modern Approach -> Russel & Norvig

Security

- [1] Network and Information Security By William Stalling
- **[2] Secure Coding in C and C++, Second Edition W

Programming Style

- **[1] Structure and Interpretation of Computer Programs (SICP)
- **[2] Concepts, Models and Techniques in Computer Programming (Haideri & Haideri)
(Considered to be modern day SICP)

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only style

- [1] clean code
- [2] great code
- [3] solid codes
- [4] programming pearls by Bentley

